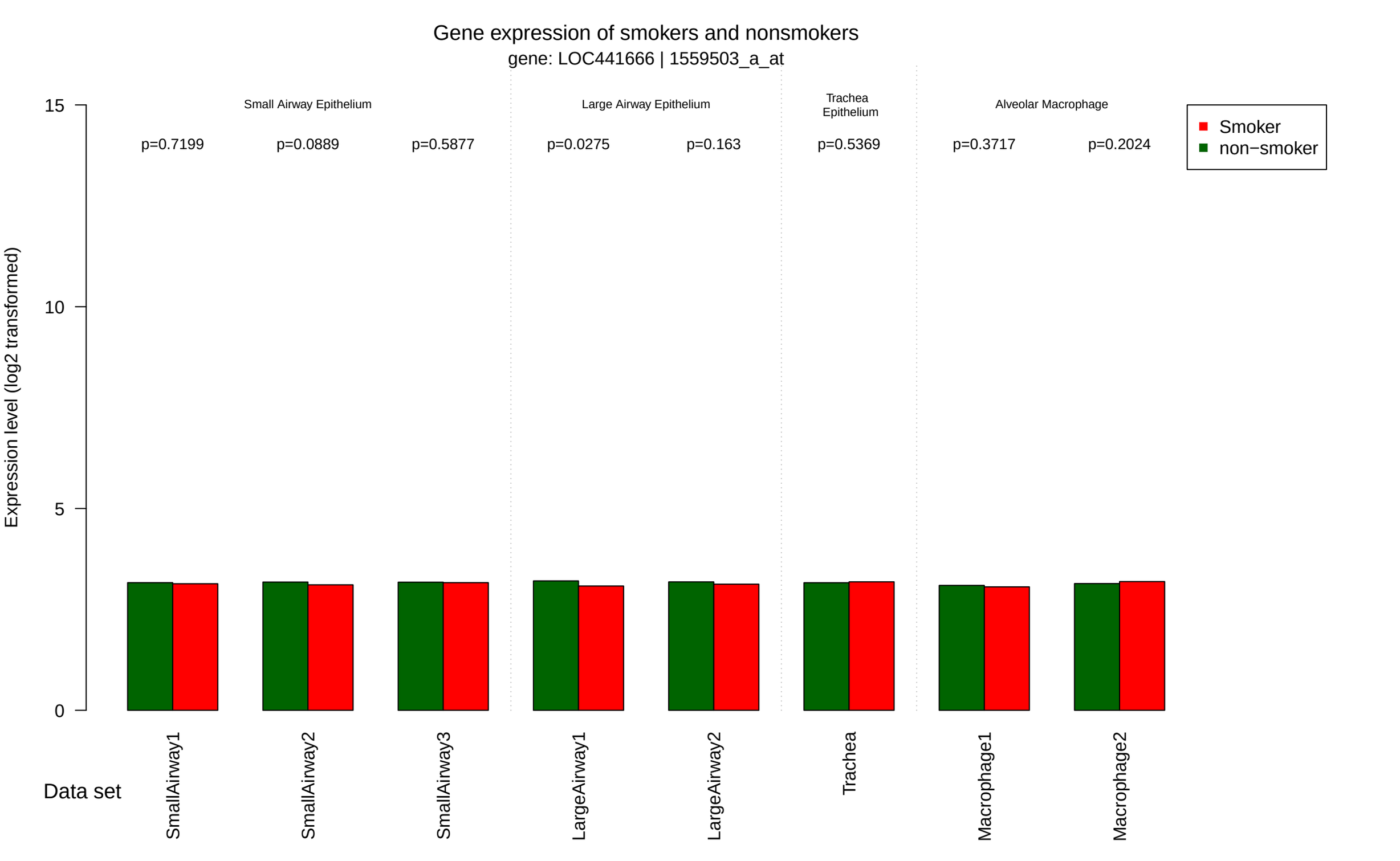
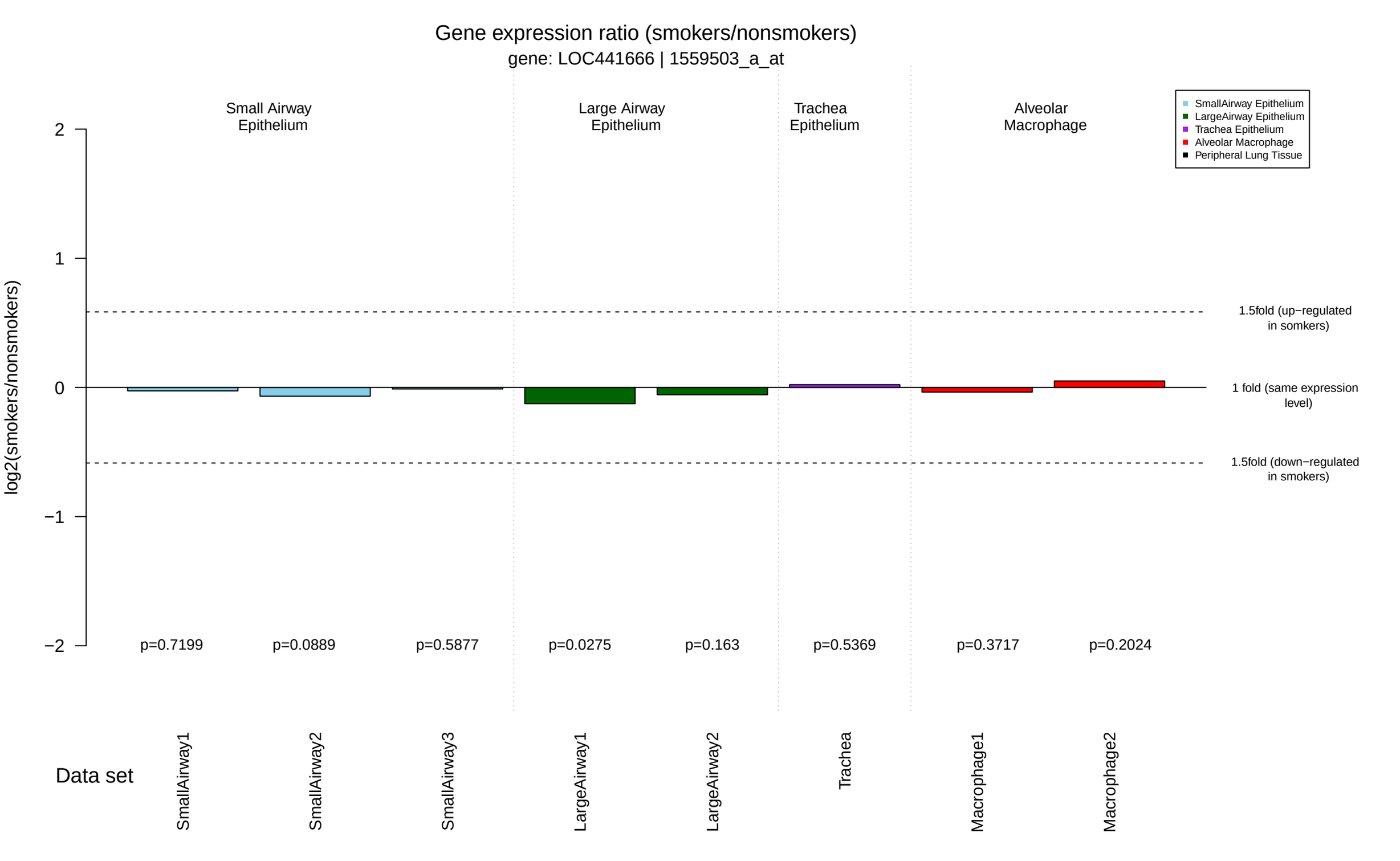
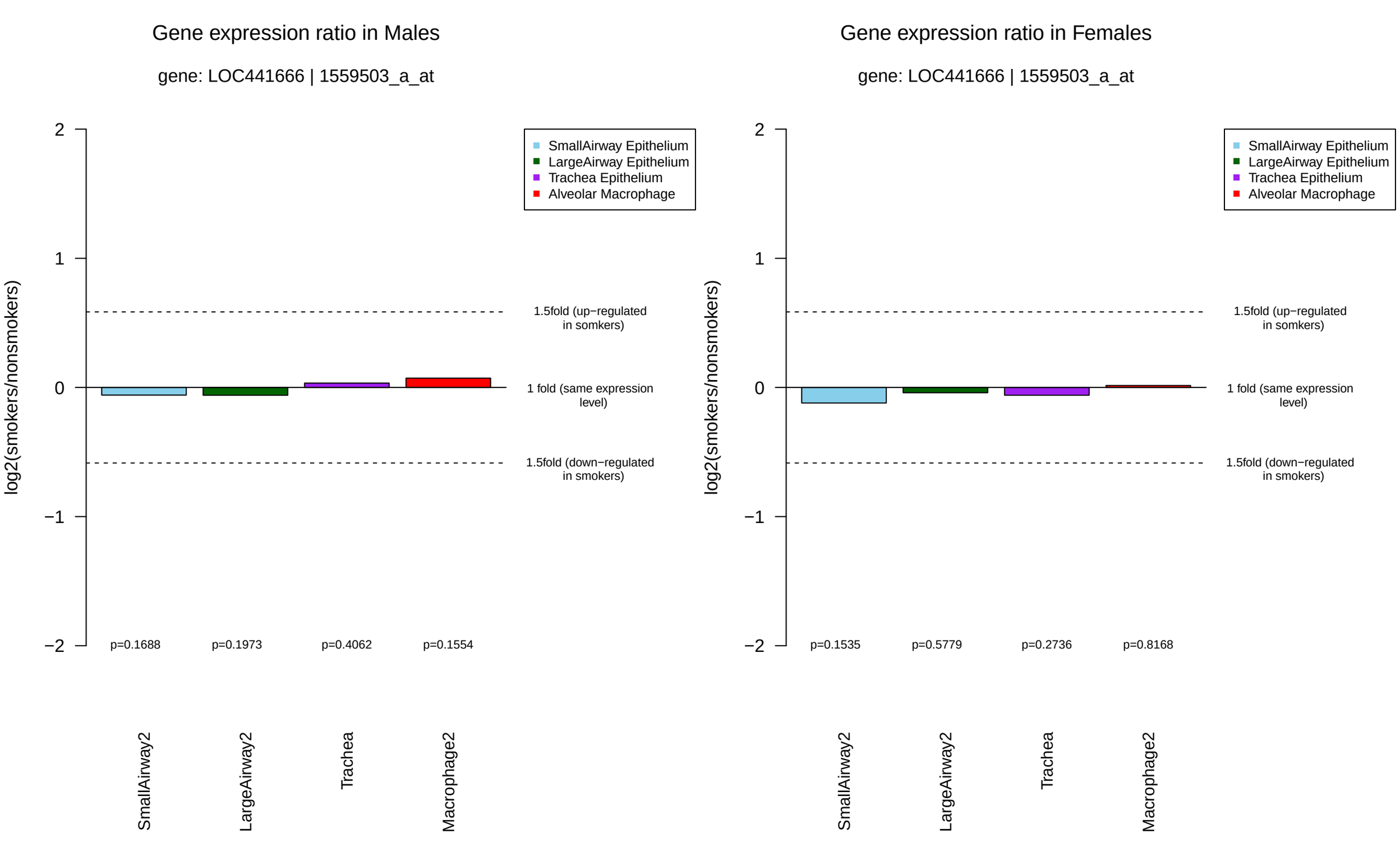
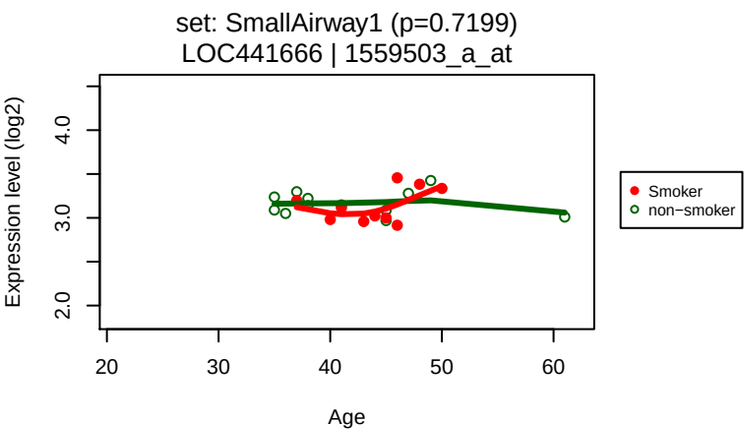
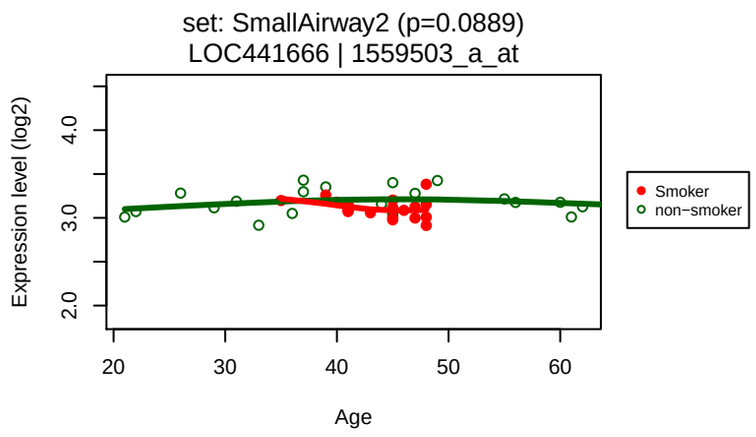
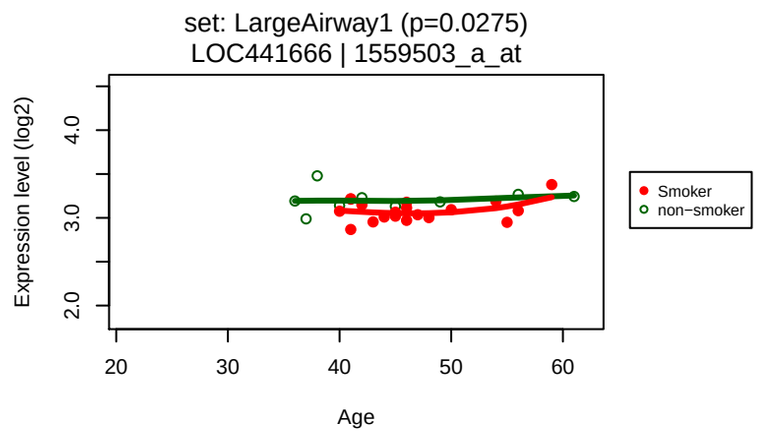

Gene expression of smokers and nonsmokers
gene: LOC441666 | 1559503_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7199
p=0.0889
p=0.5877
p=0.0275
p=0.163
p=0.5369
p=0.3717
p=0.2024
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

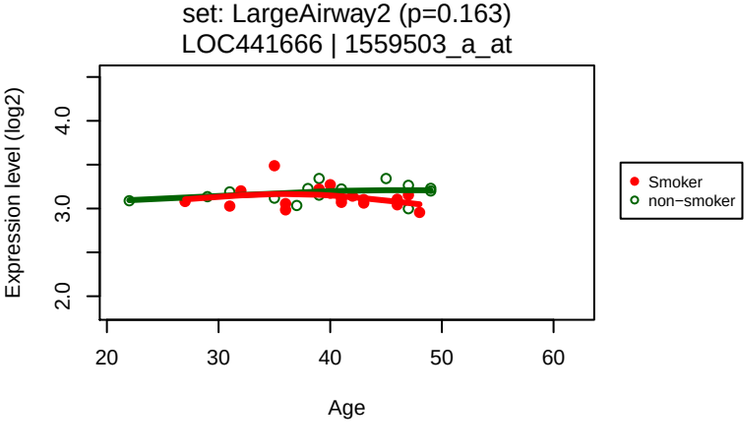

Gene expression ratio (smokers/nonsmokers)
gene: LOC441666 | 1559503_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7199
p=0.0889
p=0.5877
p=0.0275
p=0.163
p=0.5369
p=0.3717
p=0.2024
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

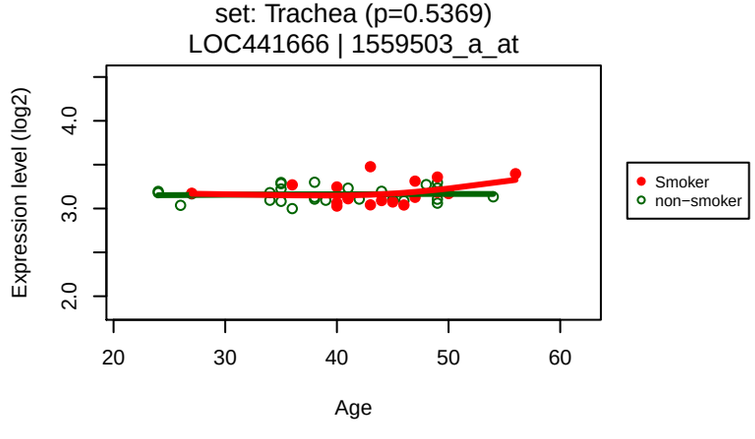

Gene expression ratio in Males
gene: LOC441666 | 1559503_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1688
p=0.1973
p=0.4062
p=0.1554
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC441666 | 1559503_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1535
p=0.5779
p=0.2736
p=0.8168
SmallAirway2
LargeAirway2
Trachea
Macrophage2

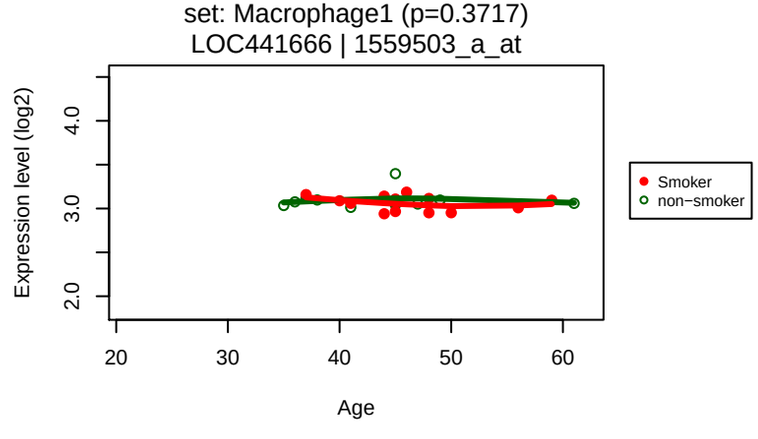

set: SmallAirway1 (p=0.7199)
LOC441666 | 1559503_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

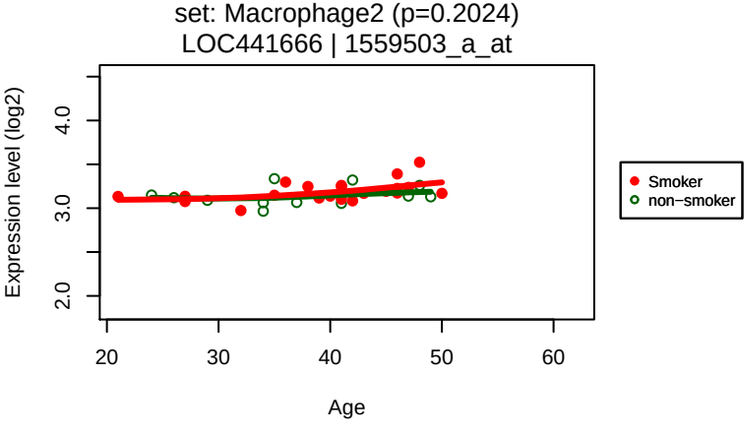

set: SmallAirway2 (p=0.0889)
LOC441666 | 1559503_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

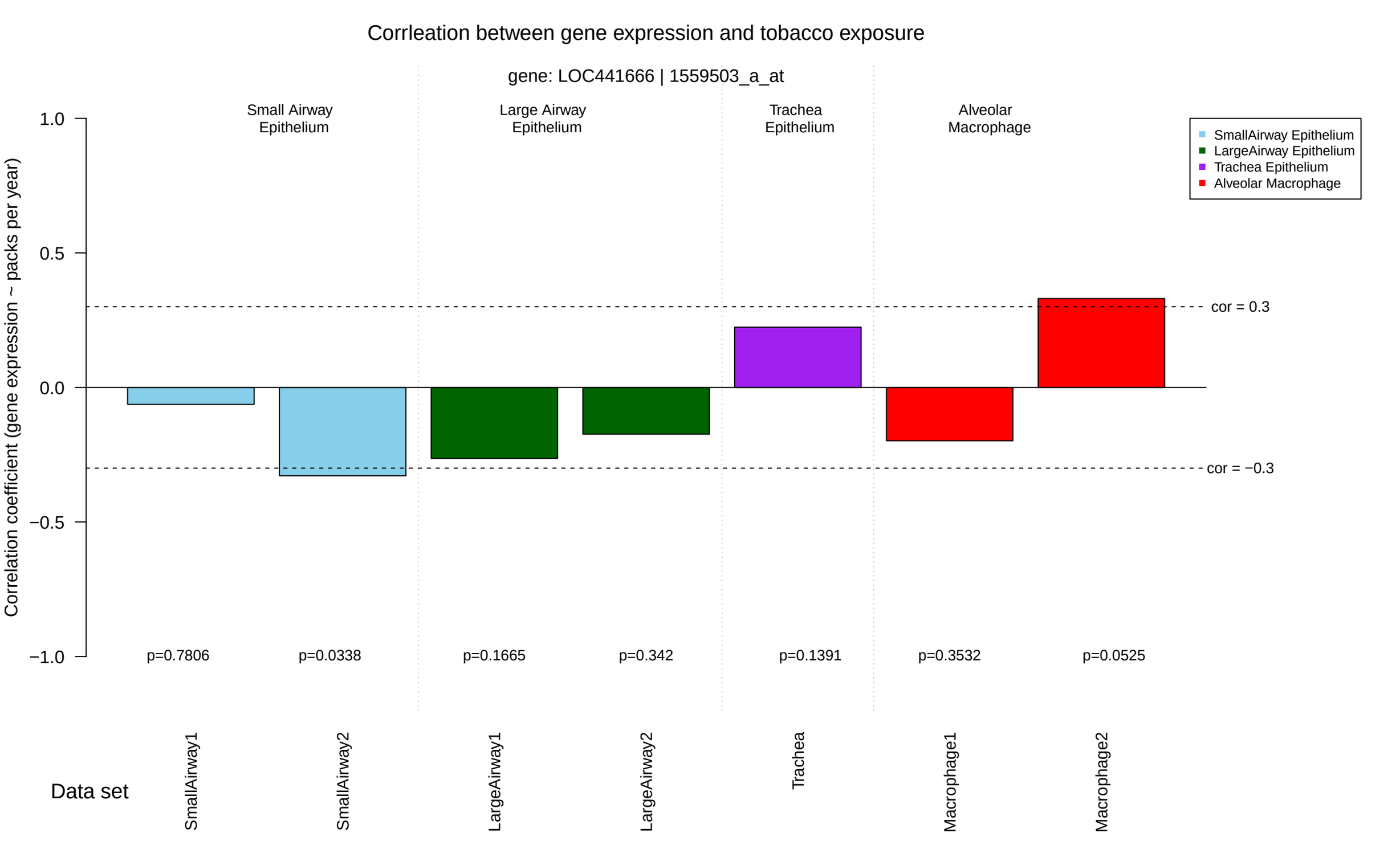

set: LargeAirway1 (p=0.0275)
LOC441666 | 1559503_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.163)
LOC441666 | 1559503_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.5369)
LOC441666 | 1559503_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3717)
LOC441666 | 1559503_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2024)
LOC441666 | 1559503_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC441666 | 1559503_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7806
p=0.0338
p=0.1665
p=0.342
p=0.1391
p=0.3532
p=0.0525
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2